

Roll No.

TCE-801

B. TECH. (CE) (EIGHT SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
EARTHQUAKE RESISTANT DESIGN OF BUILDINGS

Time : 90 Minutes

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Find out the expression for calculating response of a mass under damped free vibration in single degree of freedom system. (CO1)

OR

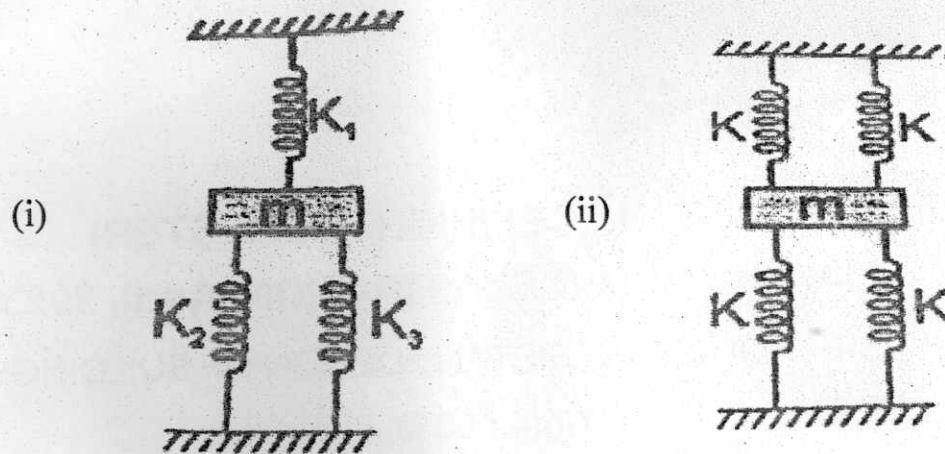
- (b) Calculate the natural frequency of system in hertz which consist of a body having mass 3 kg is attached at free end of a cantilever beam of length 2 m having cross section (150 mm × 200 mm) and Young's modulus of 210 GPa. (CO1)
2. (a) Successive amplitude of vibrations of a vibrating system as obtained under free vibrations are 0.69, 0.32, 0.19 and 0.099 units respectively. Determine the damping ratio of system. (CO1)

OR

- (b) How would you find the ratio of transmissibility of forces between a machine (having rotating mass) and its foundation ? (CO1)

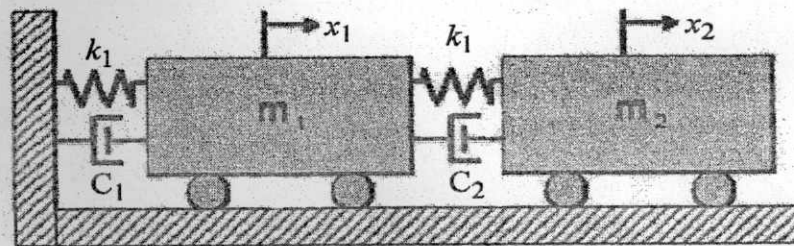
P. T. O.

3. (a) Find the effective stiffness of the springs attached in system : (CO2)



OR

- (b) Write down the equation of motion for the given multidegree of freedom system : (CO2)



4. (a) Explain elastic rebound and plate tectonic theory responsible for occurrence of earthquake. (CO3)

OR

- (b) Mention the detailed scaling of earthquake on richer magnitude scale based upon observation and discuss the relation between magnitude and intensity scale. (CO3)

(3)

5. (a) How does the frequency of a vibrating system get influenced by the excitation force provided ? What happens when resonance occurs ? (CO3)

OR

- (b) What is the main idea behind the working of a seismograph and how can it be used for design of earthquake resistant structures ? (CO3)

Roll No.

TCE-813

**B. TECH. (CIVIL ENGINEERING) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023**

HYDROPOWER ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) The power duration curve for a hydroelectric installation shows that 876×10^3 kWh of electrical energy can be obtained during the year. The plant is to be designed as a peak load plant for an annual load factor 25%. Determine the capacity of station. If the plant capacity factor is 20%, then find the reserve capacity of plant. (CO1)

OR

- (b) Predict the load of a town in year 2018 using Scheer's formula and the following data : (CO1)

Per capita load in year 2023 = 350 kWh

Population growth rate = 5% (constant for five years period)

Population in year 2023 = 50,00,000

P. T. O.

2. (a) The yearly output of a base load plant with 25 MW installed capacity is 125×10^6 kWh. The plant takes a peak of 22.5 MW. Calculate the annual load factor and capacity factor. (CO1)

OR

- (b) Discuss the relative advantages and disadvantages of the hydropower and the thermal power plant. (CO1)
3. (a) A runoff river station with an installation capacity of 15000 kW operated at 15% load factor, when it serves at peak load station. What would be the lowest discharge in the stream so, that the station may serve as a base load station? It is given that the plant efficiency is 75% when working under head of 20 m. Also calculate the maximum load factor of the plant when the discharge in the stream rises to $20 \text{ m}^3/\text{s}$. (CO1)

OR

- (b) A four way is to design to meet out the pondage requirement of a runoff river scheme. The daily flow in the river is $20 \text{ m}^3/\text{s}$. The plant is to be used as 15 hrs. peaking station. Calculate the pondage factor and storage capacity of four ways. (CO1)
4. (a) A 100 MW reversible pump-turbine has to work under a head of 400 m choose a suitable specific speed and running speed for the machine. (CO2)

OR

- (b) In a hydroelectric station, water is available at the rate of $175 \text{ m}^3/\text{s}$ under a head of 18 m the turbine run at speed of 150 r.p.m. with overall efficiency of 82%. Find the number of turbine required if they have the maximum specific speed of 460. (CO2)

(3)

5. (a) At kisneyama P-S Plant in Japan, the two reversible pump-turbines that have been used have the following rating during the pumping operation : Speed 225 r.p.m., Head – 230 m, Discharge – $86 \text{ m}^3/\text{s}$. Determine the approximate submergence necessary. (CO2)

OR

- (b) A Turbine develop 6620 kW under a head of 20 m at 130 r.p.m. Calculate the specific speed of the turbine and state the type of turbine. (CO2)